

Information Storage and Retrieval

CS418

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What is Information Retrieval?

- You have a collection of documents
 - Books, web pages, journal articles, photographs, video clips, tweets, a weather database, ...
- You have an *information need (query)*
 - *“How many species of sparrow are native to New England?”*
 - *“Find a new musician I’d enjoy listening to.”*
 - *“Is it cold outside?”*
- You want the documents that best satisfy that need



Web Search

The image displays three overlapping screenshots of web search results for the query "How many species of sparrow are native to New England?".

- Google (Top Left):** Shows the search bar with the query and a search button. Below the search bar are tabs for Web, Images, Maps, Shopping, More, and Search tools. The results section shows "About 3,790,000 results (0...". The first result is "House Sparrow - Wikipedia" with a snippet: "One of about 25 species in the House Sparrow is also native to New Zealand in 1859, and".
- Bing (Middle):** Shows the search bar with the query and a search button. Below the search bar are tabs for WEB, IMAGES, VIDEOS, MAPS, NEWS, and MORE. The results section shows "77,600 RESULTS" and "Any time". The first result is "Your Backyard Birds: House Sparrow - Welcome to New England" with a snippet: "... blog.newenglandbirdhouse.com/.../about-house-sparrow".
- Yahoo! (Bottom Right):** Shows the search bar with the query and a search button. Below the search bar are tabs for Web, Images, Video, Shopping, Maps, Blogs, and More. The results section shows "Your Backyard Birds: House Sparrow - Welcome to New England ..." with a snippet: "Interesting Facts about the House Sparrow. Get to know your backyard birds in this series of articles from New England Birdhouse.".

The Yahoo! results also include "House Sparrow - Wikipedia, the free encyclopedia" with a snippet: "One of about 25 species in the genus Passer, the House Sparrow is native to ... when birds from England were released in New ... many other animals ...".

Site-specific Search

Me sparrows

Results for **sparrows**

Top / All / People you follow

Photos · View all



E. xx @snowbalirry 3m
Harrys **sparrows** are like my favourite tattoo that he has and when they stick out of his shirt I die inside..
Expand Reply Retweet Favorite More

Evelyn Schaffer @eviefrances_ 11m
Do **sparrows** scare eagles or lions fear hares?
Expand Reply Retweet Favorite More

f Captain Jack Sparrow



Captain Jack Sparrow
21,599,309 likes · 64,516 talking about this

Like Follow

Movie
"You seem somewhat familiar. Have I threatened you before?"

About · Suggest an Edit

Highlights

Captain Jack Sparrow shared a link.
December 29, 2013

With a pirate leading the way, you can't go wrong.



Invite Your Friends to Like This Page
Type a friend's name...



Product Search

[Try Prime](#)

Jesse's Amazon.com

Today's Deals

Gift Cards

Sell

Help



Shop by Department ▾

Search

All ▾ sparrows

Go

Hello, Jesse

Your Account ▾

Try Prime ▾

 Cart ▾

Wish List ▾

Departments

Books

Bird Watching

Outdoors & Nature Reference

Ornithology

Nature & Wildlife Photography

Ecology

+ See more...

Movies & TV

Documentary

Movies

DVD

Drama

Kindle Store

Bird Watching

+ See All 32 Departments

Eligible for Free Shipping

Free Shipping by Amazon

Book Format

Paperback

Hardcover

Kindle Edition

Audible Audio Edition

Audio CD

Audio Cassette

MP3 CD

Book Language

☐ English

☐ French

☐ German

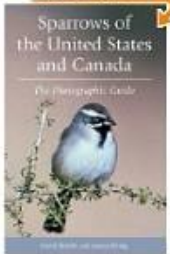
☐ Spanish

"sparrows"

Showing 1 - 16 of 58,641 Results

Choose a Department ▾ to enable sorting





Sparrows of the United States and Canada: The Photographic Guide by David Beadle and James D. Rising (Nov 30, 2001)

~~\$29.95~~ **\$19.59** Paperback 

Order in the next **4 hours** and get it by Thursday, Jan 9.

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\$16.98 new (19 offers)

\$15.60 used (21 offers)

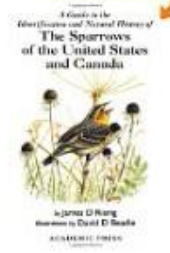
★★★★★ ☒ (24)

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A Guide to the Identification and Natural History of the Sparrows of the United States and Canada (Natural World... by James D. Rising (Sep 4, 1996)

\$28.18 new (19 offers)

\$29.97 used (23 offers)

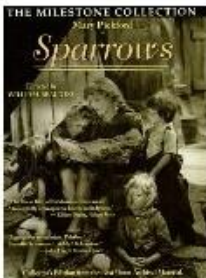
\$30.00 collectible (1 offer)

★★★★★ ☒ (3)

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Sparrows 1999 Unrated

from **\$57.90** DVD

★★★★★ ☒ (13)

Runtime: 1 hr 24 mins

Starring: Mary Pickford, Roy Stewart, et al.

Directed by: Tom McNamara and William Beaudine

Trade in this item for an Amazon.com Gift Card

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But also grouping related documents



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Happy Birthday, Mr. Dictator: Rodman sings for Kim

USA TODAY - 4 minutes ago

BEIJING - Dennis Rodman sang Happy Birthday to North Korean dictator Kim Jong Un on Wednesday before leading a team of former NBA stars for a game of "basketball diplomacy" that has been criticized by rights groups.

Opinion: [Dennis Rodman's moral responsibility](#) CNN

In Depth: [Dennis Rodman in North Korea: Has he finally gone too far?](#) Christian Science Monitor - by Peter Grier

Related [Dennis Rodman](#) » [North Korea](#) » [National Basketball Association](#) »





ABC News

Five Arrested In Murder Of Former Miss Venezuela And Her Ex-Husband

Fox News Latino - 12 minutes ago

CARACAS, Venezuela - Five unidentified suspects have been arrested for killing former Miss Venezuela and her Irish ex-husband in front of their 5-year-old daughter when they resisted a robbery on a roadside in Venezuela.



Telegraph...

Mohammed Morsi's trial delayed

Telegraph.co.uk - 1 hour ago

The Muslim Brotherhood on Wednesday demanded the release of Egypt's deposed president, Mohammed Morsi, after the interior ministry failed to bring him to court for a trial hearing citing "bad weather".



Watchdog body urges Syria to speed up chemical handover

Personalize this!

Tools to make Google News yours



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Spotlight

[Pope Francis condemns fundamentalism, urges setting an example over ...](#)

Raw Story - Jan 3, 2014

[Anti-Semitic or not, 'quenelle' gesture shows bigger issues in France \(+video\)](#)

Christian Science Monitor - 13 hours ago

[Politics|Popular Voice in the Capitol? It's the Pope's](#)

New York Times - Jan 6, 2014

[Democracy in Peril in Asia](#)

New York Times - Jan 6, 2014

[Why Iraq's Most Violent Province Is a War Zone Again](#)

TIME - Jan 4, 2014

Most Popular

[Five grilled in murder of Venezuela star Monica Spear and husband](#)

And mining the web for knowledge

Google

Web Images Maps Shopping More Search tools

David Bowie > Albums

								
The Rise and Fall of Ziggy Sta... 1972	The Next Day 2013	Aladdin Sane 1973	Hunky Dory 1971	Diamond Dogs 1974	The Man Who Sold the World 1970	Station to Station 1976	"Heroes" 1977	David Bowie 1969

[Category:David Bowie albums - Wikipedia, the free encyclopedia](#)

en.wikipedia.org/wiki/Category:David_Bowie_albums

This category contains **albums** by **David Bowie**. See also categories: **David Bowie** songs, **Tin Machine** albums, and **David Bowie** album covers ...

[David Bowie discography - Wikipedia, the free encyclopedia](#)

en.wikipedia.org/wiki/David_Bowie_discography

Jump to **Studio albums** - 1972, **The Rise and Fall of Ziggy Stardust and the Spiders from Mars**. Released: 6 June 1972; Label: RCA; Format: LP. 5, 75, 11 ...

The Singles Collection - **Best of Bowie** - **Tin Machine** - **Tin Machine II**

[David Bowie \(1967 album\) - Wikipedia, the free encyclopedia](#)

[en.wikipedia.org/wiki/David_Bowie_\(1967_album\)](http://en.wikipedia.org/wiki/David_Bowie_(1967_album))

David Bowie is the debut **album** by British musician **David Bowie**, released in 1967 on Deram Records. Its content bears little overt resemblance to the type of ...

[David Bowie - Wikipedia, the free encyclopedia](#)

en.wikipedia.org/wiki/David_Bowie

David-Bowie Chicago 2002-08-08 photoby Adam-Bielawski-cropped. ... "Starman" and the album **The Rise and Fall of Ziggy Stardust and the Spiders from Mars**. ... Bowie's latest studio album **The Next Day** was released in March 2013. **Hunky Dory** (1971) found Visconti, Bowie's producer and bassist, supplanted in both ...

[David Bowie Discography--The List of David Bowie Albums](#)

www.brianhartzoa.com/david-bowie/david-bowie-discographv.htm



David Bowie

Musician

David Robert Jones, known by his stage name David Bowie, is an English musician, singer-songwriter, producer, actor and arranger.

Wikipedia

Born: January 8, 1947 (age 67), Brixton, London, United Kingdom

Spouse: **Iman Abdulmajid** (m. 1992), **Angela Bowie** (m. 1970–1980)

Children: **Alexandria Zahra Jones**, **Duncan Jones**

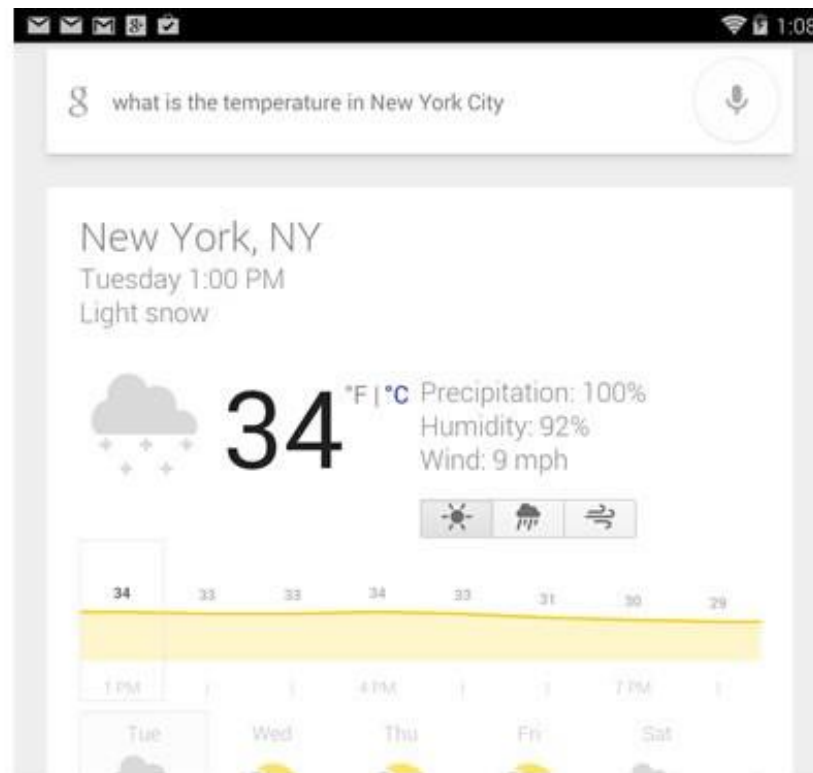
Movies: **Labyrinth**, **The Prestige**, **The Man Who Fell to Earth**, **More**

Get updates about David Bowie?

Keep me updated



And answering everyday questions



Siri and Google Now.
(Credit: Screenshot by Lance Whitney/CNET)



Course Goals

- To help you understand the fundamentals of search engines.
 - How to crawl, index, and search documents
 - How to evaluate and compare different search engines
 - How to modify search engines for specific applications
- To provide broad coverage of the major issues in information retrieval
- To take a closer look at particular applications of Information Retrieval in industry



Course Materials

- Suggested books:
 - Search Engines: Information Retrieval in Practice, by Croft, Metzler, and Strohman
 - Introduction to Information Retrieval, by Manning, Raghavan, and Schütze
 - Available for free online!



Course Topics

- Architecture of a search engine
- Data acquisition / web Crawling
- Text representation
- Information extraction
- Indexing
- Query processing
- Ranking
- Evaluation
- Classification and clustering
- Social search
- More...



A brief history of IR

- Search of digital libraries was one of the earliest tasks computers were used for.
- By the 1950s, rudimentary search systems could find documents that contained particular terms.
- Documents were ranked based on how often the specific search terms appeared in them — *term frequency* weighting



A brief history of IR

- In the 60s, new techniques were developed that treated a document as a *term vector*.
 - Using a “bag of words” model: assuming that the number of occurrences of each term matters but term order does not
 - A query can also be represented as a term vector, and the vectors can be compared to measure similarity between the document and query
- Work also started on clustering documents with similar content
- The concept of *relevance feedback* was introduced: the best few documents are assumed to be matches, and documents which are similar to them are assumed to also be relevant to the original query.
- Some of the first commercial systems appeared in the 60s, sold to companies who wanted to search their private records



A brief history of IR

- Before the Internet, search was mainly about finding documents in your own collection
 - The emphasis was largely on *recall* — making sure you find every relevant document
 - Documents were mainly text files, and did not contain references to other documents
- Just after the Internet, this was all changed
- Collection sizes jumped to *billions* of documents
 - Documents are structured in networks, providing extra relevance information, and often have other useful metadata (e.g. how many *FaceBook likes*?)
 - You can't possibly know what's in every document
 - A “document” can be pages long or just 120 characters, or could be an *image* or *video* clip, a file download, an abstract fact, or something else entirely
 - You usually care more about *precision* — making sure your first few results are relevant — because people only look at the first few results (except for when they don't...)



Challenges of IR

- Text documents are generally free-form
 - The metadata is there, but you have to find it
 - Most web pages contain lots of extra content — ads, navigation bars, comments — that might or might not be of interest
 - Spam filtering is hard
- Searching multimedia content has its own challenges
 - What are the features? How do you extract them?



Challenges of IR

- Running a query is hard
 - You have less than one second to search the full text of billions of documents to find the best ten matches
 - ...and the user only gave you two or three words
 - ...and one was misspelled, and one was “the”
 - ...and maybe throw a good relevant ad in, so you can pay the bills
- Working at web scale means massive distributed systems, sub-linear algorithms, and careful use of heuristics



Challenges of IR

- Comparing the query text to the document text and determining what is a **good match** is the core issue of information retrieval
 - Exact matching of **words is not enough**
 - Many different ways to write the same thing in a “natural language” like English
 - e.g., does a news story containing the text “bank director in Amherst steals funds” match the query “bank scandals in western mass?”
 - Some stories will be better matches than others



Relevance

- What is relevance?
- Simple (and simplistic) definition: A relevant document contains the information that a person was looking for when they submitted a query to the search engine
- Many factors influence a person's decision about what is relevant: e.g., task, context, novelty, style



Relevance

- Retrieval models define a particular view of relevance based on some idea of what users want
- Ranking algorithms used in search engines are based on retrieval models
- Most models are based on statistical properties of text rather than deep linguistic analysis
 - i.e., counting simple text features such as words instead of parsing and analyzing the sentences



Users and Information Needs

- Search evaluation is user-centered
- Keyword queries are often poor descriptions of actual information needs
- Interaction and context are important for understanding **user intent**
- Query refinement techniques such as **query expansion**, query suggestion, relevance feedback improve ranking



Research and Industry

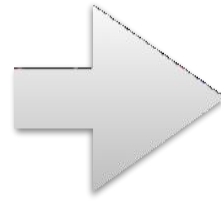
- A search engine is the practical application of information retrieval techniques to large scale text collections
- Web search engines are the best-known examples, but there are many others
- Open source search engines are important for research and development
 - e.g., **Lucene**, Lemur/Indri, Galago
- Researchers are focused on many, but not all, of the tasks that industry search engines care about



Research and Industry

Research Tasks

- Relevance
 - Effective ranking
- Evaluation
 - Testing and measuring
- Information needs
 - User interaction



Search Engines

- Performance
 - Efficient search and indexing
- Incorporating new data
 - Coverage and freshness
- Scalability
 - Growing with data and users
- Adaptability
 - Tuning for applications
- Specific problems
 - e.g. Spam



Search Engine Issues

- Performance
- Measuring and improving the efficiency of search
 - e.g., reducing **response** time, increasing **query** throughput, increasing **indexing** speed
- Indexes are data structures designed to improve search efficiency
 - Designing and implementing them are major issues for search engines



Search Engine Issues

- Dynamic data
- The “collection” for most real applications is constantly changing in terms of updates, additions, deletions
 - e.g., web pages
- Acquiring or “**crawling**” the documents is a major task
 - Typical measures are **coverage** (how much has been indexed) and **freshness** (how recently was it indexed)
- Updating the indexes while processing queries is also a design issue



Search Engine Issues

- Scalability
 - Making everything work with **millions** of users every day, and many **terabytes** of documents
 - **Distributed** processing is essential
- Adaptability
 - Changing and tuning search engine components such as ranking **algorithms**, **indexing** strategies, **interfaces** for different applications



Search Engine Issues

- Spam
- For web search, spam in all its forms is one of *the* major issues
- Affects the efficiency of search engines and, more seriously, the *effectiveness* of the results
- Proliferation of spam varieties
 - e.g. spamdexing or term spam, link spam, “optimization”
- New subfield called *adversarial IR*, since spammers are “adversaries” with different goals

